

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 2: Industry Problem Solving Task**

**COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B**

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**COURSE OUTCOME COVERED**

**CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) Assessment Tool Weightage: 40%**

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**QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A**  
**Industry Problem Solving Task**

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***Code of Conduct:***

***I \_\_Harshavardhan. RV (192472163)\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.***

***Signature of the student: HARSHAVARDHAN RV***

**Rubrics for Evaluation**

| Criteria                            | Weightage (%) | Level 4 (Exemplary)                                                          | Level 3 (Proficient)                                 | Level 2 (Developing)                           | Level 1 (Beginning)                               | Marks(100) |
|-------------------------------------|---------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|------------|
| Understanding of Industry Problem   | 20%           | Comprehensive understanding of real world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |            |
| Application of Engineering Concepts | 25%           | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |            |
| Solution Design                     | 25%           | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |            |
| Use of Tools                        | 15%           | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |            |

|                 |     |                                                                     |                                          |                                        |                                     |  |
|-----------------|-----|---------------------------------------------------------------------|------------------------------------------|----------------------------------------|-------------------------------------|--|
| Analysis        | 10% | Thorough analysis with validated results and performance evaluation | Good analysis with reasonable validation | Basic analysis with limited validation | Poor or no analysis                 |  |
| Professionalism | 5%  | Highly professional, well documented, and clearly presented work    | Good documentation and presentation      | Basic documentation with some gaps     | Poor documentation and presentation |  |

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

### 1. Traffic Surveillance Model Selection

A traffic monitoring system must:

- Accuracy  $\geq 90\%$
- Latency  $\leq 80$  ms
- Works under varying lighting

Options:

Method A: 88%, 60 ms

Method B: 91%, 85 ms

Method C: 92%, 75 ms

Compare and evaluate which method satisfies all constraints. Justify logically.

**Answer:** A traffic surveillance system is used to monitor vehicles, pedestrians, traffic violations, and road conditions in real time. For effective operation, the system must satisfy three important constraints: accuracy of at least 90%, latency of at most 80 ms, and reliable operation under varying lighting conditions.

#### Given Methods

| Method   | Accuracy | Latency |
|----------|----------|---------|
| Method A | 88%      | 60 ms   |
| Method B | 91%      | 85 ms   |
| Method C | 92%      | 75 ms   |

#### Method A

Method A provides an accuracy of 88% and a latency of 60 ms. Its latency satisfies the requirement

because 60 ms is less than 80 ms. However, its accuracy is below the required 90%.

In a traffic surveillance system, lower accuracy may result in missed vehicles, incorrect vehicle classification, or failure to identify traffic events. Therefore, Method A cannot satisfy all the system requirements.

### **Method B**

Method B provides 91% accuracy, which is greater than the required 90%. However, its latency is 85 ms, which exceeds the maximum allowed latency of 80 ms.

The higher latency can reduce the responsiveness of the traffic monitoring system. In real-time applications, delayed detection can affect traffic management and safety-related decisions.

Therefore, Method B also fails to satisfy all the requirements.

### **Method C**

Method C provides 92% accuracy and 75 ms latency.

Accuracy requirement:

$$92\% \geq 90\%$$

Latency requirement:

$$75 \text{ ms} \leq 80 \text{ ms}$$

Thus, Method C satisfies both numerical constraints. For varying lighting conditions, it should also be tested with daytime, nighttime, shadows, bright sunlight, and low-light images.

### **Comparison and Evaluation**

Method A is fast but does not provide sufficient accuracy. Method B provides sufficient accuracy but has excessive latency. Method C provides both high accuracy and low latency.

Therefore, Method C provides the best balance between detection accuracy and real-time response.

### **Conclusion**

Method C is the most suitable method because it achieves 92% accuracy and 75 ms latency, satisfying the specified accuracy and latency constraints. It should additionally be validated under different lighting conditions before deployment.

## 2. Drone Vision System Constraint

A drone system requires:

- Lightweight computation
- Real-time detection (<60 ms)
- Accuracy  $\geq 85\%$

Options:

Model A: 83%, 40 ms

Model B: 87%, 65 ms

Model C: 86%, 55 ms

Compare and evaluate the suitable model. Justify using constraints.

**Answer:**

A drone vision system requires fast and accurate object detection while operating with limited computational resources. Since drones have restrictions on battery capacity, processor capability, weight, and power consumption, the selected model must be computationally efficient and capable of real-time operation.

The requirements are:

1. Lightweight computation
2. Detection time less than 60 ms
3. Accuracy of at least 85%

### Given Models

| Model   | Accuracy | Detection Time |
|---------|----------|----------------|
| Model A | 83%      | 40 ms          |
| Model B | 87%      | 65 ms          |
| Model C | 86%      | 55 ms          |

### Model A

Model A has a detection time of 40 ms, which satisfies the real-time requirement. However, its accuracy is only 83%, which is below the required 85%.

Although it is fast and computationally attractive, insufficient accuracy can cause incorrect object detection or missed obstacles. Therefore, Model A is not suitable.

## Model B

Model B provides 87% accuracy, which satisfies the accuracy requirement. However, its detection time is 65 ms, which is greater than the required 60 ms.

For a moving drone, excessive detection delay can affect obstacle avoidance, navigation, and decision-making. Hence, Model B does not satisfy the real-time constraint.

## Model C

Model C provides 86% accuracy and a detection time of 55 ms.

Accuracy:

$$86\% \geq 85\%$$

Detection time:

$$55 \text{ ms} < 60 \text{ ms}$$

Therefore, Model C satisfies both requirements.

It also provides a reasonable balance between computational efficiency, accuracy, and real-time performance.

## Comparison and Evaluation

Model A is the fastest but does not meet the accuracy requirement. Model B provides sufficient accuracy but fails the real-time requirement. Model C satisfies both accuracy and latency constraints.

For a drone system, Model C is therefore the most practical option, provided its computational requirements are lightweight enough for the available drone hardware.

## Conclusion

**Model C is the most suitable model** because it provides 86% accuracy and 55 ms detection time. It satisfies the required accuracy and real-time constraints and provides the best balance for a resource-constrained drone vision system.

### 3. Security System Recognition

A security system must:

- High accuracy (>92%)
- Moderate computation

Options:

Traditional: 85%, low computation

Deep Learning: 94%, high computation

Hybrid: 91%, moderate computation

Compare and evaluate the best approach. Justify logically.

**Answer:**

A security recognition system is used for applications such as face recognition, access control, identity verification, and surveillance. Such systems require high accuracy because incorrect recognition can result in false alarms or unauthorized access.

The requirements are:

- Accuracy greater than 92%
- Moderate computational requirement

#### Given Approaches

| Approach      | Accuracy | Computation |
|---------------|----------|-------------|
| Traditional   | 85%      | Low         |
| Deep Learning | 94%      | High        |
| Hybrid        | 91%      | Moderate    |

#### Traditional Approach

The Traditional approach requires low computational resources, which is beneficial for systems with limited hardware.

However, its accuracy is only 85%, which is far below the required accuracy of greater than 92%.

Therefore, although it is computationally efficient, it is not suitable for a high-security recognition system.

## **Deep Learning**

Deep Learning achieves 94% accuracy, which satisfies the accuracy requirement.

It can learn complex features from large datasets and generally provides better recognition performance than traditional methods.

However, it requires high computational resources, while the system requires moderate computation. Therefore, it does not completely satisfy the requirements.

## **Hybrid Approach**

The Hybrid approach requires moderate computation, which satisfies the computational requirement.

However, its accuracy is only 91%. Since the required accuracy is greater than 92%, it fails the accuracy requirement.

## **Overall Evaluation**

Based strictly on the given values, none of the three approaches completely satisfies both requirements.

Deep Learning is the strongest candidate because it achieves 94% accuracy. Its computational cost could be reduced using techniques such as:

- Model pruning
- Quantization
- Knowledge distillation
- Lightweight network architectures
- Hardware acceleration

These techniques can reduce computational requirements while attempting to preserve high accuracy.

## **Conclusion**

No given approach completely satisfies the specified constraints. Deep Learning is the best approach to develop further because it achieves 94% accuracy, exceeding the required level. Its computational requirement should be optimized to bring it down from high to moderate. The Hybrid method has suitable computation but insufficient accuracy.

#### **4. Motion Tracking in Sports Analytics**

Requirements:

- Accurate motion tracking
- Real-time performance

Options:

Optical Flow: accurate, slow

Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

**Answer:**

Motion tracking is an important application in sports analytics. It can be used to track athletes, players, balls, body movements, speed, direction, and performance. The system must provide both accurate tracking and sufficiently fast processing.

The two available approaches are Optical Flow and Motion Estimation.

##### **Optical Flow**

Optical Flow estimates the movement of pixels between consecutive video frames. It can provide detailed information about object movement and is useful for accurate motion analysis.

Its main advantage is high accuracy. It can capture detailed motion patterns and is useful for applications where precise tracking is required.

However, Optical Flow is computationally intensive and can be relatively slow. This makes it less suitable for applications requiring strict real-time performance.

##### **Motion Estimation**

Motion Estimation determines movement between consecutive frames with lower computational complexity.

Its major advantage is high processing speed. It can provide results quickly and is therefore more suitable for real-time applications.

However, its accuracy is lower than Optical Flow. This may result in less precise tracking, especially when the movement is complex or detailed analysis is required.



### Comparison

| Factor                    | Optical Flow | Motion Estimation |
|---------------------------|--------------|-------------------|
| Accuracy                  | High         | Lower             |
| Processing speed          | Slow         | Fast              |
| Computational requirement | High         | Lower             |
| Real-time suitability     | Lower        | Higher            |
| Detailed motion analysis  | Better       | Moderate          |

### Evaluation

The requirement includes both accurate motion tracking and real-time performance. Therefore, there is a trade-off between the two methods.

For a live sports analytics system, such as real-time player tracking or live match analysis, fast processing is essential. In this situation, Motion Estimation is more practical.

For applications such as post-match analysis, where immediate results are not required, Optical Flow may be preferred because higher accuracy is more important than processing speed.

A hybrid approach could also be considered. Motion Estimation can be used for fast initial tracking, while Optical Flow can be applied selectively when detailed tracking is required.

### Conclusion

Motion Estimation is more suitable for strict real-time sports analytics because it provides faster processing. Optical Flow is preferable when maximum tracking accuracy is required and some processing delay is acceptable. Therefore, the final choice depends on the priority between real-time speed and tracking accuracy.

## 5. Face Recognition System Design

Constraints:

- Accuracy  $\geq 90\%$
- Works in low light
- Moderate computation

Options:

Traditional: 82%

Deep Learning: 93%

Hybrid: 89%

Compare and evaluate the best choice. Justify logically.

**Answer:**

Face recognition is widely used in security systems, attendance monitoring, access control, and surveillance. A reliable face recognition system must maintain high accuracy even when environmental conditions such as lighting change.

The requirements are:

- Accuracy of at least 90%
- Works effectively in low-light conditions
- Moderate computational requirement

### Given Approaches

| Approach      | Accuracy | Computation |
|---------------|----------|-------------|
| Traditional   | 82%      | Low         |
| Deep Learning | 93%      | High        |

|        |     |          |
|--------|-----|----------|
| Hybrid | 89% | Moderate |
|--------|-----|----------|

### **Traditional Approach**

Traditional face recognition methods require relatively low computational resources. This makes them suitable for systems with limited hardware.

However, the accuracy is only 82%, which is below the required 90%.

Traditional techniques can also be affected by changes in lighting, facial expressions, pose, and image quality. Therefore, this approach is not suitable for the given requirements.

### **Deep Learning**

Deep Learning achieves 93% accuracy, which satisfies the required minimum of 90%.

Deep learning models can learn complex facial features and can be trained using images captured under different lighting conditions. Therefore, they can be designed to provide better performance in low-light environments.

However, the major disadvantage is its high computational requirement, while the system requires moderate computation.

### **Hybrid Approach**

The Hybrid approach requires moderate computation, which satisfies the computational requirement.

However, its accuracy is only 89%, which is below the required 90%.

Therefore, it cannot satisfy the complete set of requirements.

### **Low-Light Requirement**

Low-light performance is an important factor in face recognition. The selected model should be trained using images from different lighting conditions. Image enhancement, noise reduction, normalization, and low-light preprocessing can also improve recognition performance.

Deep Learning is more suitable for this requirement because it can learn robust features from diverse training data.

### **Overall Evaluation**

Strictly based on the given values, none of the three approaches satisfies all three requirements.

However, Deep Learning is the strongest candidate because it achieves 93% accuracy. Its high computational requirement can potentially be reduced using:

- Model compression
- Quantization
- Pruning
- Lightweight neural network architectures
- Hardware acceleration

### **Conclusion**

Deep Learning is the best approach among the given options, because it is the only approach that exceeds the required 90% accuracy. Its computational cost must be optimized to achieve moderate computation, and it should be trained or enhanced specifically for low-light conditions. Therefore, an optimized lightweight Deep Learning model would be the most appropriate solution.